# Sahadat Hossain

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LinkedIn | GitHub | Portfolio

## Professional Summary

Cybersecurity-focused Computer Science & Telecommunication Engineering student with internship experience as a Junior Penetration Tester. Skilled in penetration testing, web application security, reconnaissance, vulnerability assessment, Linux, Python, Burp Suite, Nmap, and OWASP Top 10.

## Technical Skills

**Programming**: Python, Bash, SQL

**Operating Systems**: Kali Linux, Windows

**Security Tools:** Burp Suite, Nmap, Wireshark, Nuclei, Nikto, FFUF, Gobuster, Dirsearch
**Reconnaissance:** Subfinder, Assetfinder, Amass, Chaos, crt.sh, Findomain, SubDomainizer, Subranger SecurityTrails, ShodanX, HTTPX, PureDNS, DNSGen, Gotator, MapCIDR

**Networking:** TCP/IP, DNS, HTTP/HTTPS, Packet Analysis

**Version Control:** Git, GitHub

**Core Skills:** Penetration Testing, Web Application Security, Vulnerability Assessment, OWASP Top 10

## Professional Experience

### Junior Penetration Tester (Training Intern)

**Byte Capsule (Remote)** **August 2025 – October 2025**

- Performed web application security assessments using Burp Suite and manual testing techniques.
- Conducted reconnaissance and asset discovery using industry-standard tools such as Subfinder, Assetfinder, Amass, HTTPX, and Nmap.
- Assisted in identifying and validating security vulnerabilities using Nuclei while following OWASP Top 10 testing methodologies.
- Prepared technical vulnerability reports with findings, risk assessments, and remediation recommendations.

## Projects

**Recon Automation Framework | GitHub** *Python | Subfinder | Assetfinder | Amass | HTTPX | Naabu | Nuclei*

- Developed a modular Python-based reconnaissance automation framework for penetration testing and bug bounty workflows.
- Automated reconnaissance tasks, including **subdomain enumeration, DNS resolution, HTTP probing, port scanning, technology detection, and screenshot capture**.
- Integrated web discovery features such as **URL discovery, JavaScript analysis, directory fuzzing, virtual host discovery, WAF/CDN detection, TLS analysis, and takeover detection**.
- Designed a scalable framework with automated reporting, dashboard support, CLI improvements, plugin architecture, and performance optimization.

**Web Vulnerability Scanner | GitHub** *Python | XSS | SQLi | SSRF | SSTI | IDOR | API Security | JWT | OAuth*

- Developed a modular Python-based web vulnerability scanner for automated security testing and vulnerability assessment.
- Implemented comprehensive vulnerability detection modules covering **XSS, SQL Injection, SSRF, SSTI, XXE, LFI/RFI, IDOR, CSRF, Open Redirect, Path Traversal, and Command Injection**.
- Added advanced security testing capabilities including **API security, authentication, JWT, OAuth, GraphQL, WebSocket, CORS, Host Header Injection, and Subdomain Takeover** detection.
- Designed supporting engines for **vulnerability correlation, risk scoring, confidence scoring, false-positive reduction, threat intelligence integration, and automated security analysis.**

**SentinelSIEM | GitHub** *Python | TypeScript | SIEM | Threat Detection | MITRE ATT&CK | SOC*

- Developed a full-stack SIEM platform for centralized security monitoring, threat detection, event analysis, and security operations workflows.
- Implemented **real-time security event processing, multi-event correlation, risk scoring, and threat detection** to identify and prioritize suspicious activities.
- Integrated **MITRE ATT&CK** mapping to provide contextual analysis of detected security events and support threat investigation.
- Built **dashboard and visualization capabilities** with authentication, RBAC, session management, audit logging, observability, and automated security monitoring.

**Secure Code Review | GitHub** *Python | Flask | SQLite | Bandit | Semgrep*

- Developed a secure Flask-based web application demonstrating modern secure coding practices and secure application design.
- Implemented security features including password hashing, SQL injection prevention, Role-Based Access Control (RBAC), secure file upload, and session-based authentication.
- Performed static application security testing (SAST) using Bandit and Semgrep, analyzing 18 source files with 151 security rules.
- Applied OWASP secure coding principles and generated security analysis reports with remediation.

**Basic Network Sniffer | GitHub** *Python | Scapy | Wireshark | Pytest*

- Developed a Python-based network packet sniffer using Scapy to capture and analyze live network traffic.
- Implemented protocol analysis, Berkeley Packet Filters (BPF), real-time traffic statistics, and packet inspection for TCP, UDP, ICMP, and ARP protocols.
- Implemented packet export in CSV, JSON, and PCAP formats for automated analysis and Wireshark compatibility.
- Integrated automated testing, logging, and GitHub Actions CI to improve code reliability and maintainability.

**Network Intrusion Detection System | GitHub** *Python | Snort 3 | Kali Linux | Pytest*

- Developed a Snort 3–based Network Intrusion Detection System (NIDS) for real-time network traffic monitoring and threat detection.
- Created custom Snort rules to detect ICMP, HTTP, SSH, FTP, and TCP SYN scan activities, with automated alert analysis.
- Generated structured security reports in TXT, CSV, and JSON formats for intrusion analysis.
- Integrated automated testing and GitHub Actions CI to improve code quality and project reliability.

## Education

**B.Sc. in Computer Science and Telecommunication Engineering** 2022 – 2026

Noakhali Science and Technology University (NSTU), Noakhali, Bangladesh.

CGPA: 3.40 / 4.00

## Certifications

- Ethical Hacking Professional (EHP) – Byte Capsule
- Blue Team Junior Analyst Pathway – Security Blue Team (BTLO)
- Advanced Real Bug Bounty Case Studies – Volume 1 – Udemy
- Certified by micro1 – Outstanding Performance in AI Interview - micro1